

## 1. Data Link Layer (DLL) Foundations & Framing

Module 01

- **DLL Core Responsibilities:** Framing, Flow Control, Error Control, Physical Addressing (MAC), and Channel Access Control (MAC Sublayer).
- **LLC (802.2) vs MAC:** LLC handles upper protocol multiplexing & flow control (medium independent). MAC handles channel access, framing & contention.
- **Framing Methods:**
  - *Character Count:* Frame header carries total byte count. Vulnerable: 1 corrupted count bit desynchronizes all subsequent frames!
  - *Byte / Character Stuffing:* Flag byte ('0x7E') delimits frame. If Flag occurs in payload, prefix with 'ESC' ('0x1B'). If ESC in payload, prefix with 'ESC'.
  - *Bit Stuffing (5-Ones Rule):* Flag = '01111110'. Sender automatically stuffs a '0' after five consecutive '1's ('111110'). Receiver unstuffs it.

Bit Stuffing: 011111101111110 -> 011111(0)1011111(0)10

## 2. Pure ALOHA vs Slotted ALOHA

Module 02

- **Pure ALOHA (Continuous):** Station transmits immediately whenever data is ready.
  - Vulnerable Time:  $T_{vu1n} = 2 \cdot T_{fr}$
  - Poisson Throughput:  $S = G \cdot e^{(-2G)}$
  - Max Throughput:  $S_{max} = 1 / (2e) \approx 18.4\%$  at  $G = 0.5$ .
- **Slotted ALOHA (Discrete Clock Slots):** Station can only transmit at beginning of slot (Slot duration =  $T_{fr}$ ).
  - Vulnerable Time:  $T_{vu1n} = 1 \cdot T_{fr}$  (Halved!)
  - Poisson Throughput:  $S = G \cdot e^{(-G)}$
  - Max Throughput:  $S_{max} = 1 / e \approx 36.8\%$  at  $G = 1.0$  (Doubled!).

AKTU Poisson:  $P(k) = (G^k \cdot e^{(-G)}) / k!$  |  $P(0) = e^{(-G)}$  (1 slot)

## 3. Carrier Sense Multiple Access (CSMA) & Persistence Strategies

Module 03

### 1-Persistent CSMA:

- Continuously senses channel.
- If Idle: Transmits immediately ( $p=1$ ).
- If Busy: Keeps continuous listening.
- High collision if 2 stations wait for busy channel to free!

### Non-Persistent CSMA:

- Senses channel once.
- If Idle: Transmits immediately.
- If Busy: Waits random backoff time before sensing again.
- Reduces collisions; wastes channel idle time. Peak ~0.8.

### p-Persistent CSMA:

- Used in slotted channels.
- If Idle: Transmits with prob  $p$ ; defers to next slot with prob  $1 - p$ .
- Optimal:  $N \cdot p < 1$  to prevent collisions.

Collision Window:  $t$  in  $[0, T_p]$  due to propagation delay. Carrier sensing cannot see ahead of wavefront!

## 4. CSMA with Collision Detection (CSMA/CD)

Module 04

- **Mechanism:** "Listen While Talk". Transceiver checks voltage:  $V_{chan} > 2 \cdot V_{tx}$  indicates collision.
- **Collision Action:** Immediately abort frame + Broadcast 32/48-bit **Jam Signal** + Increment  $k$  (collision count).
- **Binary Exponential Backoff (BEB):** Wait  $R \cdot Slot\_Time$  where  $0 \leq R \leq 2^k - 1$  (Max  $k=10$ ; Abort at  $k=16$ ).
- **Core Derivation (64-Byte Frame):**

$T_t \geq 2 \cdot T_p \implies L_{min} = 2 \cdot T_p \cdot Bandwidth = 512 \text{ bits} = 64 \text{ Bytes}$

## 5. CSMA with Collision Avoidance (CSMA/CA)

Module 05

- **Why CSMA/CD Fails in Wireless:**
  - Dynamic Range: Transmit power drowns received power (Tx blinds Rx).
  - Hidden Station Problem: A & C cannot hear each other; collide at B.
  - Exposed Station: C refrains from sending to D due to hearing A-B.
- **Interframe Spaces (IFS):**  $SIFS < PIFS < DIFS < EIFS$  (Shortest IFS = Highest Priority; SIFS for ACK/CTS).
- **4-Way RTS/CTS Handshake:** RTS carries duration; CTS echoes it. Stations overhearing CTS set **NAV** countdown timer (Virtual Sensing).

$NAV = SIFS + T_{CTS} + SIFS + T_{DATA} + SIFS + T_{ACK}$

6. Controlled Access & CDMA Channelization

Module 06

Controlled Access Methods:

- Reservation:** N bit reservation interval precedes N data frames.
- Polling:** Primary-Secondary architecture. Poll (asks data), Select (sends data). Single point of failure at Primary.
- Token Passing:** Special Token frame circulating in logical ring. Station holding token transmits for Token Holding Time (THT). Zero collisions!

CDMA & Orthogonal Walsh Codes:

- Walsh Recursive Construction:  $W_{2N} = [W_N, W_N; W_N, -W_N]$
- Properties:  $W_i \cdot W_i = N$  and  $W_i \cdot W_j = 0$  ( $i \neq j$ )
- Data Encoding: Bit 0 = -1, Bit 1 = +1, Silent = 0. Transmitted:  $S_i = d_i \cdot W_i$ .
- Channel Signal:  $C = \sum S_i$  | Receiver Decoding:  $d_i = (C \cdot W_i) / N$

Example:  $C = [1, 1, -1, 3], W2 = [1, -1, 1, -1] \rightarrow C \cdot W2 = 1 - 1 - 1 - 3 = -4 \rightarrow d2 = -4/4 = -1$  (Bit 0 Decoded!)

7. Flow Control Foundations & Stop-and-Wait ARQ Protocol

Module 07

- Flow Control:** Prevents fast sender from overflowing slow receiver buffer.
- Error Control (ARQ):** Adds Timer + Retransmission + Sequence Numbers.
- 1-Bit Sequence Numbering (0 & 1):** Prevents duplicate frame acceptance when ACK is lost! Receiver ACK n expects frame n.
- Piggybacking:** Outgoing ACK combined inside outgoing data frame header.

Total Time  $T_{total} = T_t + 2 \cdot T_p$  (Neglecting Tack, Tproc)

Parameter  $a = T_p / T_t = (d / v) / (L / \text{Bandwidth})$

Efficiency  $\eta = T_t / (T_t + 2 \cdot T_p) = 1 / (1 + 2a)$  | Throughput  $= \eta \cdot B$

Physical Impact of 'a': If  $a \gg 1$  (Satellite link,  $a=250$ ):  $\eta$  drops to  $1 / (1 + 500) = 0.2\%$ ! Motivates Sliding Window Pipelining.

8. Sliding Window Protocols: Go-Back-N vs Selective Repeat ARQ

Module 08 & 09

| Parameter / Feature    | Stop-and-Wait ARQ                           | Go-Back-N (GBN) ARQ                                         | Selective Repeat (SR) ARQ                                                  |
|------------------------|---------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------|
| Sender Window (Ws)     | 1                                           | $2^m - 1$                                                   | $2^{(m - 1)}$                                                              |
| Receiver Window (Wr)   | 1                                           | 1 (Strictly sequential)                                     | $2^{(m - 1)}$ (Out-of-order buffer!)                                       |
| Window Sizing Proof    | $Ws + Wr \leq 2^m \rightarrow 1 + 1 \leq 2$ | $Ws + 1 \leq 2^m \implies Ws \leq 2^m - 1$                  | $Ws + Wr \leq 2^m \implies 2 \cdot Ws \leq 2^m \implies Ws \leq 2^{(m-1)}$ |
| Out-of-Order Buffering | No                                          | No (Subsequent frames discarded!)                           | Yes (Stored in reassembly buffer)                                          |
| Acknowledgment Type    | Individual ACK                              | Cumulative ACK (ACK n confirms up to n-1)                   | Individual ACK + NAK (Selective Reject)                                    |
| Retransmission Scope   | 1 frame                                     | All N frames in window (Go-Back penalty)                    | Only the damaged / lost frame                                              |
| Number of Timers       | 1 timer                                     | 1 timer (For oldest unacknowledged frame Sf)                | Multiple independent timers (1 per frame)                                  |
| Efficiency ( $\eta$ )  | $1 / (1 + 2a)$                              | $Ws / (1 + 2a)$ (for $Ws < 1+2a$ ; 100% if $Ws \geq 1+2a$ ) | $Ws / (1 + 2a)$ (High utilization even on lossy link)                      |

GBN 100% Efficiency Condition:  $Ws \geq 1 + 2a \implies m \geq \lceil \log_2(2 + 2a) \rceil$

Why  $Ws \leq 2^m - 1$ ? If  $Ws = 2^m$  and all ACKs lost, sender retransmits old window; receiver cannot distinguish old duplicate from new frame!

9. Error Detection: Parity, Checksum & CRC

Module 10

- Error Types:** Single-Bit vs Burst Error (Length = first to last flipped bit index).
- Parity:** 1D parity fails on even-bit flips. 2D Parity detects all 1, 2, 3-bit errors; corrects 1-bit error via row-column intersection.
- Internet Checksum:** 16-bit 1's complement sum. Wrap-around carry added to LSB. Final result complemented. Receiver sum must be all 1s.
- Cyclic Redundancy Check (CRC):** Modulo-2 XOR division.
  - Degree of  $G(x)$  is  $r$ . Append  $r$  zeros to data.
  - Divide by  $G(x)$  using Modulo-2 arithmetic.
  - Remainder  $R$  is CRC (FCS). Codeword = Data + R.
  - Receiver checks if  $\text{Codeword} \bmod G(x) == 0$ .

CRC-32: Used in 802.3 Ethernet FCS. Detects 100% burst errors  $\leq 32$  bits!

10. Error Correction & Hamming Code

Module 11

- Hamming Distance ( $d_{min}$ ):**
  - To detect  $d$  errors:  $d_{min} \geq d + 1$
  - To correct  $t$  errors:  $d_{min} \geq 2t + 1$
- Redundancy Formula:**  $2^r \geq m + r + 1$  (For  $m=4$  bits data  $\rightarrow r=3$  bits  $\rightarrow$  Hamming (7, 4) Code).
- Bit Positions:** Parity at powers of 2 (1, 2, 4). Data at 3, 5, 6, 7.

$p1 = d1 \oplus d2 \oplus d4 \mid p2 = d1 \oplus d3 \oplus d4 \mid p4 = d2 \oplus d3 \oplus d4$
- Syndrome Word ( $S4\ S2\ S1$ ):**

$S1 = b1 \oplus b3 \oplus b5 \oplus b7 \mid S2 = b2 \oplus b3 \oplus b6 \oplus b7 \mid S4 = b4 \oplus b5 \oplus b6 \oplus b7$

Value of ( $S4\ S2\ S1$ ) directly gives the erroneous bit index! Invert to correct.

11. IEEE 802.3 Ethernet Frame & Standards Evolution

Module 12

| Preamble         | SFD               | Dest MAC      | Src MAC       | Length/Type        | Data Payload (LLC PDU)             | FCS (CRC-32) |
|------------------|-------------------|---------------|---------------|--------------------|------------------------------------|--------------|
| 7 Bytes (1010..) | 1 Byte (10101011) | 6 Bytes (48b) | 6 Bytes (48b) | 2 Bytes (Type/Len) | 46 to 1500 Bytes (Padding if <46B) | 4 Bytes      |

- Frame Limits:** Minimum = 64 Bytes (512 bits); Maximum = 1518 Bytes.
- MAC Address (48 bits):** First 24 bits = OUI (Vendor ID); Last 24 bits = NIC serial. Broadcast = `FF:FF:FF:FF:FF:FF`.
- Ethernet Family:** 10Base5 (Thick), 10Base2 (Thin), 10Base-T (Twisted pair). Fast Ethernet (100 Mbps, 802.3u). Gigabit Ethernet (1 Gbps, 802.3z). 10G Ethernet (802.3ae, Full-Duplex only, No CSMA/CD).

12. Token Ring (802.5) & FDDI

Module 13

- IEEE 802.5 Token Ring:** Physical star, logical ring via MSAU. Token = SD (1B) + AC (1B) + ED (1B). Active Monitor clears orphan frames & lost tokens. Early Token Release in 16 Mbps.
- FDDI (100 Mbps Optical Fiber):**
  - Dual Counter-Rotating Ring (Primary CW + Secondary CCW).
  - Self-Healing: Cable cut triggers wrap-around into 1 single ring!
  - Timed Token Protocol:  $THT = \max(0, TTRT - TRT)$ .

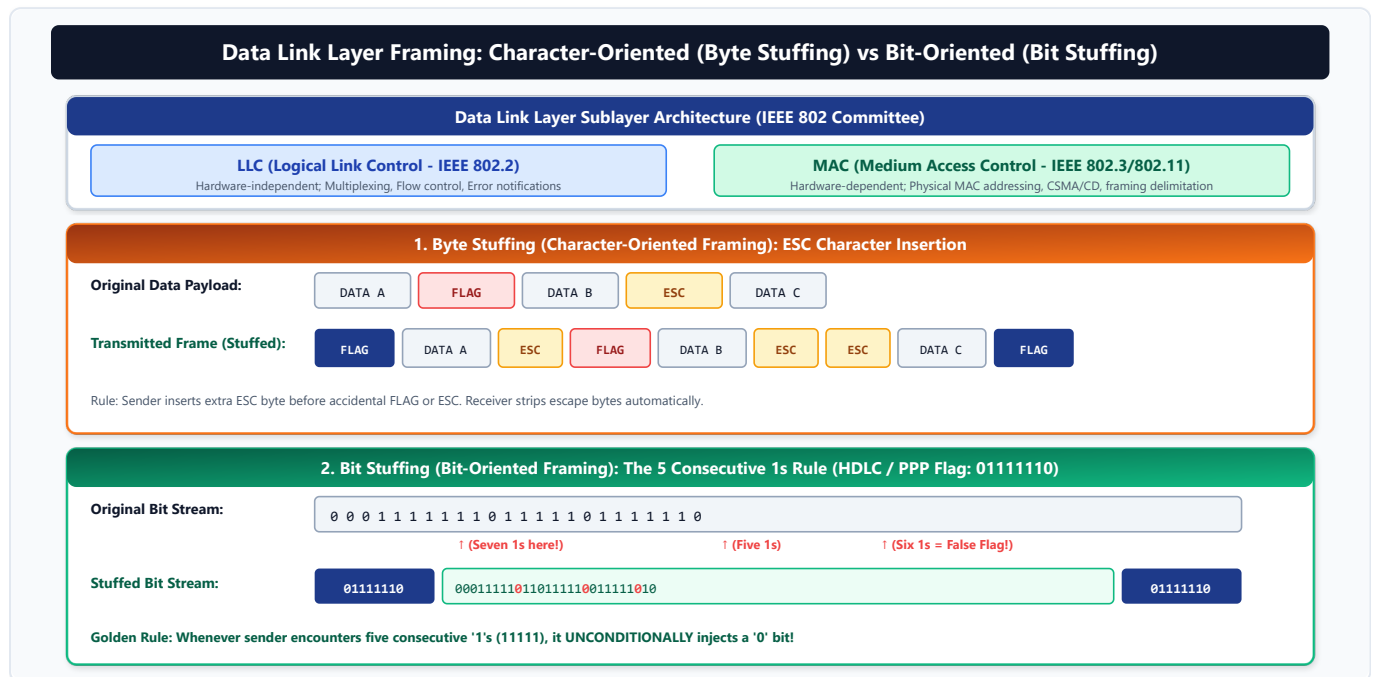
13. Bridges, Switches & STP (802.1D)

Module 14

- Collision vs Broadcast Domains:** Switch splits collision domains (1 per port); preserves 1 broadcast domain.
- Transparent Bridge Learning:** Learn (Source MAC), Forward/Filter (Dest MAC), Flood (Unknown/Broadcast).
- Spanning Tree Protocol (STP 802.1D):** Prevents broadcast storms & loops.
  - 1. Elect Root Bridge (Lowest Bridge ID: Priority + MAC).
  - 2. Select Root Port (RP) per non-root bridge (Lowest path cost).
  - 3. Select Designated Port (DP) per segment.
  - 4. Place redundant ports in **BLOCKING** state!

# Module 01: Data Link Layer Foundations & Framing Techniques

**AKTU BCS603 Core Topic:** LLC vs MAC sublayers, Character Count vulnerability, Byte Stuffing escape sequences, and Bit Stuffing 5-ones rule.



## 1. Comparative Framing Strategies

| Framing Technique | Demarcation Mechanism                                   | Vulnerability / Overhead                                                | Standard Protocol                    |
|-------------------|---------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------|
| Character Count   | Header integer stores frame length                      | Single bit count error desynchronizes all subsequent frames             | Early framing experiments (Obsolete) |
| Byte Stuffing     | Flag bytes with ESC byte prefix                         | Tied to 8-bit character boundaries; high byte overhead                  | BISYNC, PPP (Byte mode)              |
| Bit Stuffing      | Flag pattern 01111110 with 0-bit injection after 5 ones | Negligible bit overhead (~1 bit per 32 bits on average); arbitrary data | HDLC, SDLC, PPP, Frame Relay         |

## 2. Golden Rule of Bit Stuffing

**Sender Rule:** Scan bit stream. Whenever five consecutive 1s (11111) are transmitted, **UNCONDITIONALLY** stuff a 0.

**Receiver Rule:** If five consecutive 1s are received:

- If next bit is 0: Destuff (strip) it!
- If next bit is 1 followed by 0: Valid **FLAG** detected!
- If next bit is 1 followed by 1: Error / Link Abort signal!

# Module 02: Random Access Protocols: Pure & Slotted ALOHA

**AKTU BCS603 Focus:** Contention resolution, Vulnerable time derivations, throughput equations, Poisson traffic model, and solved numerical problems.

### Random Access MAC Protocols: Pure ALOHA vs Slotted ALOHA

#### 1. Pure ALOHA (Unsynchronized / Continuous)

**Vulnerable Time =  $2 \cdot T_{fr}$**

**Vulnerable Window:**  $[t - T_{fr}, t + T_{fr}]$

**$S = G \cdot e^{-2G}$**

Max Throughput  $S_{max} = 1 / (2e) \approx 0.184$  (18.4%) at  $G = 0.5$

- Station transmits immediately whenever data is ready.
- Highest collision probability; completely uncoordinated.

**Even a 1-bit overlap destroys both transmitting frames!**

#### 2. Slotted ALOHA (Discrete Clock Synchronized)

**Vulnerable Time =  $1 \cdot T_{fr}$**

**Vulnerable Window:** Exactly 1 Slot ( $T_{fr}$ )

**$S = G \cdot e^{-G}$**

Max Throughput  $S_{max} = 1 / e \approx 0.368$  (36.8%) at  $G = 1.0$

- Time divided into discrete slots equal to frame duration  $T_{fr}$ .
- Station **MUST** wait for beginning of next slot to transmit.

**Doubles throughput capacity compared to Pure ALOHA!**

#### ALOHA Mathematical Evaluation & Poisson Distribution Metrics

**Traffic Load (G):** Average number of frames generated by the whole system per frame transmission time ( $T_{fr}$ ).

**Probability of k Frames:**  $P(k) = (G^k \cdot e^{-G}) / k!$  Probability of zero frame collisions  $P(0) = e^{-G}$  for 1 slot;  $e^{-2G}$  for 2 slots.

**AKTU Numerical Formula:** Idle Slots Probability =  $e^{-G}$ . If 10% slots idle:  $e^{-G} = 0.10 \rightarrow G = \ln(10) \approx 2.3$ . Throughput  $S = G \cdot e^{-G} = 2.3 \times 0.1 = 0.23$ .

## 1. ALOHA Mathematical Summary

| Protocol      | Vulnerable Time ( $V_{t\$}$ ) | Throughput Function   | Peak Load ( $G^{*}$ ) | Maximum Efficiency        |
|---------------|-------------------------------|-----------------------|-----------------------|---------------------------|
| Pure ALOHA    | $2 \cdot T_{fr}$              | $S = G \cdot e^{-2G}$ | $G = 0.5$             | <b>18.4%</b> ( $1 / 2e$ ) |
| Slotted ALOHA | $1 \cdot T_{fr}$              | $S = G \cdot e^{-G}$  | $G = 1.0$             | <b>36.8%</b> ( $1 / e$ )  |

## 2. Why Slotted ALOHA Outperforms Pure ALOHA

Pure ALOHA me agar koi dusri station kisi frame ke beech me bhi transmit kar de, to collision ho jata hai. Slotted ALOHA stations ko time boundaries par constrain karta hai: transmissions sirf slot start par ho sakti hain. Isse collision window aadhi ho jati hai ( $2T_{fr}$  o  $T_{fr}$ ), aur channel throughput seedhe double ho jata hai!

**AKTU Exam Shortcut:** Agar question me Poochha jaye ki Pure ALOHA me  $G=0.5$  par throughput kya hai:  $S = 0.5 \cdot e^{-1} = 0.5 / 2.718 = 0.184$  (18.4%). Calculator par  $e^{-x}$  function use karna na bhoolein!

# Module 03: Carrier Sense Multiple Access (CSMA) & Persistence

**AKTU BCS603 Focus:** Carrier sensing principles, 1-Persistent vs Non-Persistent vs p-Persistent strategies, and Propagation Delay collision window analysis.

### CSMA Philosophy ("Listen Before Talk") & 3 Persistence Strategies

#### 1. 1-Persistent CSMA

##### Greedy Transmission

Sense channel continuously

- If Idle: Transmit immediately ( $P = 1$ ).
- If Busy: Keep sensing continuously.

The very instant line becomes idle, transmit!

##### High Collision Danger!

If 2+ nodes wait during busy period, both fire together when idle!

- Zero idle delay when line free
- Used in Classic IEEE 802.3 Ethernet

#### 2. Non-Persistent CSMA

##### Polite Random Wait

Relinquishes immediate sensing

- If Idle: Transmit immediately.
- If Busy: Back off random time!

Do NOT continuously monitor line.

##### Drastically Reduces Collisions!

Different random waits prevent simultaneous firing

- Channel remains idle unnecessarily
- Lower channel efficiency at low load

#### 3. p-Persistent CSMA

##### Probabilistic Slotted Compromise

Balances collision & idle wait

- If Idle: Transmit with probability  $p$ .
- With prob  $(1 - p)$ : Wait for next slot.

In next slot: Repeat test if still idle.

##### Mathematical Optimum

If  $N$  nodes ready, setting  $p = 1/N$  avoids collisions

- Optimal trade-off under heavy load
- Used in IEEE 802.11 (CSMA/CA)

### Why Collisions Still Occur in CSMA: The Propagation Delay ( $T_p$ ) Window

**Root Cause of CSMA Collisions:** Signals take finite propagation time ( $T_p = d/v$ ) to travel down the wire.

**Collision Scenario:** Host A senses channel (idle) and transmits at  $t=0$ . Before A's signal reaches distant Host B, B senses channel at  $t = T_p - \epsilon$ . B also sees channel as "idle" and transmits.

**Vulnerable Window:** CSMA Vulnerable Time = Exactly Propagation Delay ( $T_p$ ). (ALOHA was  $2T_{fr}$  or  $T_{fr}$ !)

## 1. Comparative Strategy Flow

- **1-Persistent:** "Aggressive". Channel idle  $\rightarrow$  Transmit immediately. Channel busy  $\rightarrow$  Keep listening, transmit at first instant of idle. *Pros:* Zero wasted channel time. *Cons:* High collision when busy period ends.
- **Non-Persistent:** "Polite". Channel idle  $\rightarrow$  Transmit immediately. Channel busy  $\rightarrow$  Wait random timer before re-sensing. *Pros:* Collision probability drops drastically. *Cons:* Channel sits idle while stations are in backoff sleep.
- **p-Persistent:** "Optimal Compromise". Slotted channel. If idle, transmit with probability  $p$ . With  $(1-p)$ , defer to next slot. Balances throughput and collision rate.

## 2. Throughput Comparison Curves (Kleinrock-Tobagi Model)

Channel load  $G$  ke badhne par throughput  $S$ :

Throughput  $S$

The graph shows the relationship between Throughput ( $S$ ) on the y-axis and Offered Load ( $G$ ) on the x-axis. The curves are as follows:

- Non-Persistent:** Reaches peak  $\sim 0.8$ . The curve starts at (0,0), rises to a peak of approximately 0.8 at  $G \approx 0.4$ , and then declines.
- 1-Persistent:** Peaks  $\sim 0.53$ , drops under heavy load. The curve starts at (0,0), rises to a peak of approximately 0.53 at  $G \approx 0.25$ , and then declines.
- Slotted ALOHA:** Peaks 0.368. The curve is a symmetric parabola starting at (0,0), peaking at  $S = 0.368$  when  $G = 0.368$ , and returning to zero at  $G = 0.736$ .
- Pure ALOHA:** Peaks 0.184. The curve is a symmetric parabola starting at (0,0), peaking at  $S = 0.184$  when  $G = 0.184$ , and returning to zero at  $G = 0.368$ .

**AKTU Exam Question:** "Discuss different carrier sense protocols. How are they different from collision protocols?"  
me 1-Persistent, Non-Persistent aur p-Persistent ka flowchart aur  $T_p$  collision window explanation zaroor de!

# Module 04: CSMA with Collision Detection (CSMA/CD)

**AKTU BCS603 Exam Core:** "Listen While Talk" mechanism,  $T_t \geq 2T_p$  derivation, Ethernet 64-byte minimum frame size origin, and Binary Exponential Backoff.

### CSMA/CD: "Listen While Talk", Condition $T_t \geq 2T_p$ & Minimum Frame Size Derivation

#### 1. Collision Detection Operation

- Core Principle: "Listen While Transmit"**  
Station continuously compares transmitted signal with received voltage.

Normal Signal:  $V_{channel} \approx V_{trans}$

Collision Detected:  $V_{channel} > 2 \cdot V_{trans}$  (Peak spike!)
- Action on Collision Detection:**
  - Immediately **ABORT** transmission (Saves bandwidth!).
  - Broadcast a 32/48-bit **JAM SIGNAL** to warn all stations.
  - Increment collision counter  $k$  ( $k = k + 1$ ).
  - Execute Binary Exponential Backoff algorithm.
  - If  $k > 15$ : Give up (Report transmission error to upper layer).

Saves immense channel time compared to sending full broken frame!

#### 2. Mathematical Condition: $T_t \geq 2 \cdot T_p$

$$T_t \geq 2 \cdot T_p$$

Transmission Delay MUST be at least Twice Propagation Delay!

- The Worst-Case Collision Detection Scenario:**
  - Host A transmits at  $t = 0$ .
  - At  $t = T_p$ , signal reaches far-end Host B.
  - B transmits; collision occurs at  $t \approx T_p$ .
  - Collision noise takes another  $T_p$  to travel back to Host A.
  - Host A detects collision at  $t = 2 \cdot T_p$  (Round Trip Time)!**

$$L_{min} = 2 \cdot T_p \cdot \text{Bandwidth (Bits)}$$

If frame finishes transmitting before  $2T_p$ , sender CANNOT detect collision!

### Binary Exponential Backoff Algorithm (BEB) in Ethernet (IEEE 802.3)

**Backoff Formula:**  
 $\text{Waiting Time} = R \cdot \text{Slot\_Time}$   
where  $R$  is chosen uniformly at random from range:  $0 \leq R \leq 2^k - 1$

**Range Scaling:**  
1st Collision ( $k=1$ ):  $R \in \{0, 1\}$ . 2nd Collision ( $k=2$ ):  $R \in \{0, 1, 2, 3\}$ . 10th Collision ( $k=10$ ):  $R \in \{0 \dots 1023\}$ .

**Ethernet Slot Time:**  
In 10 Mbps Ethernet: Slot Time =  $51.2 \mu s$  = 512 bit times = **64 Bytes (Minimum Frame Size!)**

## 1. Minimum Frame Size Formula Derivation

A sender must not finish transmitting before the worst-case collision signal returns:

$$T_t \geq 2 \cdot T_p \implies \frac{L}{B} \geq 2 \cdot \frac{d}{v} \implies L_{\text{min}} = 2 \cdot d \cdot B$$

**Critical Engineering Implication:** Agar network bandwidth badhai jaye (e.g. 10 Mbps to 100 Mbps to 1 Gbps), to same minimum frame size maintain karne ke liye maximum network cable distance  $d$  ko proportionally **kam** karna padta hai! Isliye Gigabit Ethernet me collision domain distance sirf 100 meters tak limited hota hai.

## 2. Binary Exponential Backoff (BEB) Summary

| Collision Count (\$k\$) | Contention Window Range (\$2^k - 1\$) | Possible Values for \$R\$            | Collision Probability (2 nodes) |
|-------------------------|---------------------------------------|--------------------------------------|---------------------------------|
| 1                       | 0 to 1                                | {0, 1}                               | 0.50 (50%)                      |
| 2                       | 0 to 3                                | {0, 1, 2, 3}                         | 0.25 (25%)                      |
| 3                       | 0 to 7                                | {0, 1, 2, 3, 4, 5, 6, 7}             | 0.125 (12.5%)                   |
| 10 (Max cap)            | 0 to 1023                             | 1024 discrete choices                | ~0.00098                        |
| > 16                    | Abort                                 | Frame dropped; Link failure reported | —                               |

# Module 05: CSMA with Collision Avoidance (CSMA/CA) & Wireless LANs

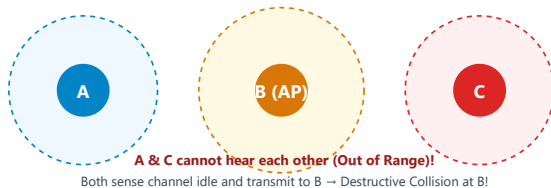
## Unit 2: Data Link Layer & MAC Sublayer

## IEEE 802.11 Standards

In-depth architectural analysis of collision avoidance in wireless networks, solving the Hidden and Exposed Station problems via SIFS/DIFS timers, RTS/CTS handshake, and NAV virtual carrier sensing.

### CSMA/CA Architecture: Collision Avoidance, IFS & RTS/CTS Handshake (IEEE 802.11)

#### 1. Why CSMA/CD Fails in Wireless (Hidden Stations)

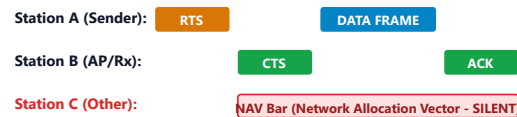


##### • Hardware Limitation in RF Transceivers:

Station's own transmission power drowns out received signals (Dynamic Range).  
Collision detection during transmission is impossible in standard Wi-Fi.

**Solution: Avoid collisions beforehand using IFS & Virtual Sensing!**

#### 2. RTS / CTS Virtual Sensing with NAV



##### • The 4-Way Handshake Steps:

1. A sends RTS (contains duration of entire upcoming transaction).
2. B responds with CTS (echoes remaining duration to all neighbors including C).
3. C hears CTS → Sets NAV timer → Suppresses transmission until ACK completes.
4. A transmits Data safely without any risk of collision from Station C!

• **SIFS < PIFS < DIFS ensures highest priority for control packets (ACK, CTS).**

SIFS (Shortest) = CTS / ACK | DIFS (Distributed) = New Data Contention

### CSMA/CA Protocol Mechanics & Interframe Space (IFS) Hierarchy

**IFS Hierarchy:** SIFS (Shortest) < PIFS (PCF Mode) < DIFS (DCF Mode) < EIFS (Extended / Error)

**Contention Window:** Station senses idle channel → Waits DIFS → Chooses random Backoff from [0, CW]. CW doubles on collision:  $CW_{min} = 15$ ,  $CW_{max} = 1023$ .

**Exposed Station:** Station B transmits to A. Station C wants to transmit to D (away from B). Senses B and unnecessarily halts. RTS/CTS solves this cleanly.

**NAV Mechanism:** Virtual Carrier Sensing: Stations do not waste battery continuously sniffing RF; hardware timer counts down NAV to zero before sensing.

## 1. Why CSMA/CD Fails in Wireless Networks

In wired Ethernet, collision detection is performed by sensing voltage spikes ( $V_{channel} > 2 \cdot V_{trans}$ ). In wireless media, radio signal power attenuates rapidly with distance ( $1/r^2$ ). Due to transceiver dynamic range limitations, a station's powerful transmitter blinds its sensitive receiver amplifier, making real-time collision detection virtually impossible.

## 2. Hidden and Exposed Station Dilemma

- **Hidden Station Problem:** Stations A and C cannot detect each other's RF signals due to physical distance or obstacles. Both sense the channel as idle and transmit to intermediate Access Point B simultaneously, resulting in catastrophic collision at station B.
- **Exposed Station Problem:** Station C unnecessarily refrains from transmitting to D because it senses active transmission between A and B, despite D being far outside B's interference range.

### 3. 4-Way RTS/CTS Handshake & NAV Virtual Sensing

---

#### NAV Duration Equation

$$\text{Duration\_RTS} = \text{SIFS} + \text{T\_CTS} + \text{SIFS} + \text{T\_DATA} + \text{SIFS} + \text{T\_ACK}$$

$$\text{NAV\_CTS} = \text{SIFS} + \text{T\_DATA} + \text{SIFS} + \text{T\_ACK}$$

Every station overhearing a CTS sets its **Network Allocation Vector (NAV)** countdown timer to the duration value, silencing transmissions until the complete data-ACK cycle finishes.

# Module 06: Controlled Access & Channelization Protocols

## Unit 2: Data Link Layer & MAC Sublayer

## Collision-Free Protocols & CDMA

Comprehensive coverage of collision-free access (Reservation, Polling, Token Passing) and channelization methods (FDMA, TDMA, CDMA) with Walsh matrix mathematical proofs and solved numericals.

### Controlled Access & Channelization: Reservation, Polling, Token Passing & CDMA Walsh Codes

#### 1. Controlled Access: Collision-Free Methods

##### A. Polling (Primary-Secondary Master/Slave):



##### B. Token Passing (Logical Ring Architecture):



Station holding TOKEN transmits for Token Hold Time (THT).

##### Guaranteed Zero Collisions & Deterministic Delay!

Trade-off: High overhead of polling/token management at low load.

#### 2. CDMA & Orthogonal Walsh Codes

##### Walsh Matrix Generation (N = 4):

$W1 = [+1, +1, +1, +1]$   $W2 = [+1, -1, +1, -1]$   
 $W3 = [+1, +1, -1, -1]$   $W4 = [+1, -1, -1, +1]$   
Property:  $W_i \cdot W_i = N = 4$  AND  $W_i \cdot W_j = 0$  (Orthogonal!)

##### • Encoding & Transmission:

Data Bit 0  $\rightarrow -1$  | Data Bit 1  $\rightarrow +1$  | Silent  $\rightarrow 0$

Transmitted Vector:  $S_i = d_i \cdot W_i$

Composite Channel Signal:  $C = S1 + S2 + S3 + S4$

##### • Receiver Decoding (Dot Product):

Decoded Data  $d_i = (C \cdot W_i) / N$

### Channelization Architecture: FDMA vs TDMA vs CDMA Comparison & Solved Example

**FDMA:** Bandwidth divided into dedicated frequency bands. Requires Guard Bands. Continuous transmission.

**TDMA:** Entire bandwidth shared in discrete time slots. Requires synchronization & Guard Times. Burst transmission.

**CDMA:** Stations transmit simultaneously across the entire bandwidth using unique orthogonal codes. No time slots!

**AKTU Example:** If S1 sends  $d1=+1$ , S2 sends  $d2=-1$ , S3 silent ( $d3=0$ ), S4 sends  $d4=+1$ :

$$C = (+1)W1 + (-1)W2 + (0)W3 + (+1)W4 = [1, 1, 1, 1] + [-1, 1, -1, 1] + [1, -1, -1, 1] = [1, 1, -1, 3]$$

## 1. Controlled Access vs Contention Protocols

Unlike contention methods where stations collide, controlled access guarantees deterministic, collision-free access. Stations either reserve time slots beforehand (Reservation), respond to master inquiries (Polling), or pass a logical permission token (Token Ring/Bus).

## 2. CDMA & Orthogonal Walsh Codes

Code Division Multiple Access allows all stations to transmit simultaneously across the entire bandwidth. Each station is assigned a unique, mutually orthogonal chip sequence generated via Walsh matrices.

### CDMA Encoding & Decoding Formulas

Transmitted Signal:  $S_i = d_i \cdot W_i$

Composite Signal:  $C = \sum S_i$

Receiver Decoding:  $d_i = (C \cdot W_i) / N$

# Module 07: Flow Control Foundations & Stop-and-Wait ARQ Protocol

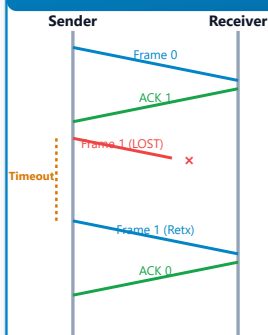
## Unit 2: Data Link Layer & MAC Sublayer

## Flow Control & Error Control

Rigorous study of Stop-and-Wait ARQ protocol mechanics, timer timeouts, 1-bit sequence numbering, mathematical efficiency derivation, and satellite link numericals.

### Stop-and-Wait ARQ: Protocol Timing, Failure Recovery & Efficiency Derivation

#### 1. Protocol Operation & Failure Scenarios



##### • Key Protocol Rules:

1. Sender sends 1 frame, starts Timer.
2. Receiver accepts, sends ACK for next expected frame (ACK 1 expects Frame 1).
3. If timer expires: Retransmit frame.

##### • Why 1-bit Sequence Numbers?

Handles Duplicate Frames when ACK is lost!  
Sequence numbers toggle: 0 and 1.  
Window Size Sender = 1, Receiver = 1.

#### 2. Efficiency & Utilization Derivation

$$\text{Total Cycle Time} = T_t + 2 \cdot T_p$$

(Assuming ACK transmission time & processing delay are negligible)

$$\eta = T_t / (T_t + 2T_p) = 1 / (1 + 2a)$$

where  $a = T_p / T_t$

##### Throughput / Useful Bandwidth:

$$\text{Throughput} = \eta \cdot \text{Bandwidth} = L / (T_t + 2T_p)$$

For large  $T_p$  (Satellite links,  $a \gg 1$ ):  $\eta$  drops below 1%!

#### AKTU Core Numerical: Stop-and-Wait ARQ on Satellite Links (Impact of Parameter 'a')

**Problem Data:** Bandwidth  $B = 1 \text{ Mbps}$  ( $10^6 \text{ bps}$ ), Frame Length  $L = 1000 \text{ bits}$ , Propagation Delay  $T_p = 250 \text{ ms}$  ( $0.25 \text{ s}$ ).

**Step 1 ( $T_t$ ):**  $T_t = L / B = 1000 / 10^6 = 1 \text{ ms}$  ( $0.001 \text{ s}$ )

**Step 2 (Ratio  $a$ ):**  $a = T_p / T_t = 250 \text{ ms} / 1 \text{ ms} = 250$

**Step 3 (Efficiency):**  $\eta = 1 / (1 + 2 \cdot 250) = 1 / 501 \approx 0.00199$  ( $0.199\%$ ) | **Throughput** =  $0.00199 \times 1 \text{ Mbps} = 1.99 \text{ kbps}$

Sender spends 99.8% of time sitting idle waiting for ACK! This motivates Sliding Window Pipelining.

## 1. Stop-and-Wait ARQ Protocol Mechanics

Stop-and-Wait ARQ enforces strict sequential flow control: the transmitter sends exactly one frame and starts a timeout timer. It pauses transmission until a valid Acknowledgment (ACK) arrives. 1-bit alternating sequence numbering (0 and 1) prevents duplicate frame delivery caused by lost ACKs.

## 2. Efficiency Derivation: The $1/(1+2a)$ Law

### Fundamental Stop-and-Wait Efficiency Formula

$$a = T_p / T_t = (d / v) / (L / B)$$

$$\eta = T_t / (T_t + 2T_p) = 1 / (1 + 2a)$$

$$\text{Throughput} = \eta \cdot \text{Bandwidth}$$

For high bandwidth-delay product links (such as satellite communication where  $a = 250$ ), link efficiency collapses to less than 0.2%, compelling the development of pipelined Sliding Window protocols.

# Module 08: Sliding Window Protocols: Go-Back-N ARQ Architecture

## Unit 2: Data Link Layer & MAC Sublayer

## Pipelined ARQ Protocols

Deep mathematical dive into Go-Back-N ARQ: sliding window pointer mechanics, cumulative acknowledgments, formal proof for window sizing  $W_S = 2^m - 1$ , and numerical derivations for 100% efficiency.

### Sliding Window Protocols: Go-Back-N (GBN) ARQ Architecture & Window Mechanics

#### 1. Window Pointers & Buffer Mechanics

Sender Window ( $m = 3 \text{ bits} \rightarrow W_S = 2^3 - 1 = 7$ ):



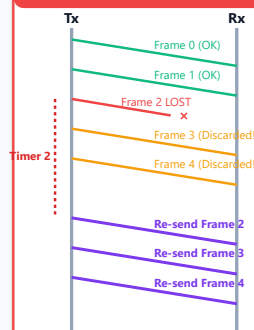
Receiver Window (Strictly  $W_R = 1$ ):



**Receiver has NO BUFFER for out-of-order frames!**

If Frame 2 is lost, arriving Frames 3, 4, 5 are DISCARDED unconditionally!

#### 2. Error Recovery: Frame Loss & Retransmit-N



##### The Go-Back-N Penalty:

Even though Frames 3 & 4 arrived without noise, receiver rejected them because Frame 2 was missing!

**Sender must GO BACK and re-send ALL N outstanding frames!**

Wastes enormous bandwidth on noisy high-speed links.

### Mathematical Foundation: Window Size Proof ( $W_S < 2^m$ ) & Pipelined Efficiency

#### Why $W_S = 2^m - 1$ ?

If  $W_S = 2^m$  (e.g.,  $m=2$ ,  $W_S=4$ , sequence {0,1,2,3}). If sender sends frames 0,1,2,3 and all ACKs are lost, receiver expects Frame 0 of NEW round. When sender retransmits old Frame 0, receiver CANNOT tell if it's new or old!

#### Efficiency Formula:

$\eta = W_S / (1 + 2a)$  for  $W_S < (1 + 2a)$  |  $\eta = 100\%$  for  $W_S \geq (1 + 2a)$

#### Optimal Window:

To achieve 100% utilization: Window Size  $W_S \geq 1 + 2a = 1 + 2(Tp/Tt)$ . Pipelining fills the transmission pipe!

## 1. Pipelining & Go-Back-N Mechanics

Pipelining overcomes the single-frame waiting penalty of Stop-and-Wait by maintaining a window of outstanding unacknowledged frames ( $W_S$ ). Receiver window is strictly  $W_R = 1$ , enforcing sequential in-order acceptance without out-of-order buffering.

## 2. Formal Proof of $W_S = 2^m - 1$

If  $W_S = 2^m$ , a complete loss of an entire burst of ACKs forces retransmissions of old frames that wrap around modulo  $2^m$ , making it mathematically impossible for the receiver to distinguish duplicate retransmitted frames from brand new frames of the next cycle.

### Go-Back-N Mathematical Governing Equations

Sender Window:  $W_S = 2^m - 1$  | Receiver Window:  $W_R = 1$

Efficiency:  $\eta = W_S / (1 + 2a)$  [for  $W_S < 1 + 2a$ ]

Optimal Window for 100%  $\eta$ :  $W_S \geq 1 + 2a$

# Module 09: Sliding Window Protocols:

## Selective Repeat ARQ Architecture

### Unit 2: Data Link Layer & MAC Sublayer

### Out-of-Order Buffering & NAK Mechanics

In-depth architectural analysis of Selective Repeat ARQ, receiver reassembly buffer logic, NAK selective rejection, and formal mathematical derivation of window sizing  $W_S = W_R = 2^{m-1}$ .

#### Selective Repeat ARQ: Independent Buffering, NAK Retransmission & Window Sizing Proof

##### 1. Window Architecture ( $m = 3 \text{ bits} \rightarrow W_S = W_R = 4$ )

Sender Window ( $W_S = 2^{m-1} = 4$ ):



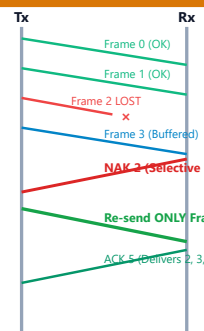
Receiver Window ( $W_R = 4$ , Has Out-of-Order Buffer):



##### • Key Architectural Advantage:

Frames 3 & 4 are STORED in Receiver Buffer, NOT discarded!  
Sender ONLY retransmits the missing Frame 2 (Zero wastel).

##### 2. Timeline: NAK 2 & Selective Retransmission



##### • Selective Repeat Merits:

1. Receiver buffers out-of-order frames; no unnecessary discards.
2. Immediate NAK triggers quick retransmission before timer expiry.

##### • Trade-off / Complexity:

Requires independent timer per frame and sorting reassembly buffers at the receiver side.

#### Mathematical Proof: Why $W_S = W_R = 2^{m-1}$ ? (AKTU Core Exam Derivation)

**Proof ( $W_S + W_R \leq 2^m$ ):**

For no overlap between old and new sequence numbers:  $\text{Window\_Sender} + \text{Window\_Receiver} \leq \text{Total Sequences} = 2^m$ .

Since receiver must buffer up to  $W_S$  frames to avoid buffer overflow, we choose  $W_S = W_R$ . Therefore:  $2 \cdot W_S \leq 2^m \rightarrow W_S \leq 2^{m-1}$ .

**Counter-Example ( $W_S > 2^{m-1}$ ):**

If  $m=2$  (sequences 0,1,2,3) and  $W_S=W_R=3$ : Sender transmits 0,1,2. Receiver accepts all and advances window to {3, 0, 1}.

If all ACKs are lost, sender retransmits old Frame 0. Receiver window contains 0 (as NEW frame)! Overlap disaster occurs!

**Master Rule:**

**Stop-and-Wait:**  $W_S=1, W_R=1$  | **Go-Back-N:**  $W_S=2^m - 1, W_R=1$  | **Selective Repeat:**  $W_S = W_R = 2^{m-1}$

## 1. Selective Repeat vs Go-Back-N

While Go-Back-N discards all subsequent frames upon encountering a gap, Selective Repeat accepts and buffers out-of-order frames ( $W_R = 2^{m-1}$ ). It selectively requests retransmission of only the missing frame via NAK frames.

## 2. Formal Proof of $W_S = W_R = 2^{m-1}$

## Window Boundary Constraint

$$W_{\text{Sender}} + W_{\text{Receiver}} \leq 2^m$$

$$\text{Since } W_S = W_R: 2 \cdot W_S \leq 2^m \rightarrow W_S \leq 2^{(m-1)}$$

If  $W_S > 2^{m-1}$ , duplicate frames from an unacknowledged old window will map directly into the advance receiver window, producing undetectable data corruption.

# Module 10: Error Detection: Parity, Checksum & Cyclic Redundancy Check (CRC)

## Unit 2: Data Link Layer & MAC Sublayer

## Error Control & Polynomial Mathematics

Mathematical foundations of transmission error detection: 2D parity, 1's complement Internet checksum, Modulo-2 polynomial division (CRC), and error catching proofs.

### Error Detection: Cyclic Redundancy Check (CRC) Polynomial Division & Checksum

#### 1. CRC Modulo-2 Binary Division (XOR Arithmetic)

Data: 100100 (6 bits) | Divisor G(x): 1101 (degree r = 3 zeros appended)

```
Divisor: 1101 ) 100100 000 ( Appended r=3 zeros )
               1101
               ---
               01000 (XOR: 1^1=0, 0^1=1, 0^0=0, 1^1=0)
                1101
                ---
                01010
                 1101
                 ---
                 01110
                  1101
                  ---
                  0011 → Remainder CRC = 001
```

Transmitted Codeword = Data + CRC = 100100 001

#### 2. Parity & Internet Checksum Methods

##### Two-Dimensional (2D) Parity:

Data organized into a matrix of rows and columns.

Adds row parity bits AND a column parity byte.

**Detects all 1, 2, 3-bit errors! Corrects any single-bit error!**

##### Internet Checksum (16-bit 1's Complement):

1. Split message into 16-bit words.

2. Add all words using 1's complement (wrap-around carry!).

3. Checksum = 1's complement of sum. Receiver sum must be all 1s.

##### Receiver Verification in CRC:

Receiver divides incoming codeword (100100001) by G(x) (1101).

**If Remainder == 0 → ACCEPT! If Remainder != 0 → ERROR DETECTED!**

#### Standard CRC Polynomials & Burst Error Detection Capabilities

**CRC-16:**  $G(x) = x^{16} + x^{15} + x^2 + 1$

Used in HDLC and USB. Appends 16 check bits.

**CRC-32:**  $G(x) = x^{32} + x^{26} + x^{23} + \dots + 1$

**Standard in IEEE 802.3 Ethernet FCS! Detects 100% burst errors  $\leq 32$  bits!**

##### Burst Error Definition:

Length of burst is measured from first corrupted bit to last corrupted bit (even if intermediate bits are intact).

##### CRC Detection Proof:

If polynomial has at least two terms: detects all single bits. If  $(x+1)$  is a factor: detects all odd numbers of bit errors.

## 1. CRC Modulo-2 Arithmetic

Cyclic Redundancy Check relies on polynomial algebra over Galois Field GF(2). Bit addition and subtraction are performed via carry-less XOR operations. Appending  $r$  zeros followed by division yields the  $r$ -bit Frame Check Sequence (FCS).

### CRC Operational Formula

Appended Message:  $M(x) \cdot x^r$

$M(x) \cdot x^r = Q(x) \cdot G(x) + R(x)$

Transmitted Codeword  $T(x) = M(x) \cdot x^r + R(x)$

Receiver Check:  $T(x) \bmod G(x) == 0$  (Valid)

# Module 11: Error Correction & Hamming Code Architecture

Unit 2: Data Link Layer & MAC Sublayer

Forward Error Correction (FEC)

Mathematical principles of error correction: Hamming distance theorems, redundancy inequality  $2^r \geq m + r + 1$ , parity placement at powers of 2, and syndrome word error location decoding.

1. Bit Positions & Parity Check Sets (7-Bit Codeword)

Pos 1

Pos 2

Pos 3

Pos 4

Pos 5

Pos 6

Pos 7

p1

p2

d1

p4

d2

d3

d4

001

010

011

100

101

110

111

• Parity Bit Mathematical Coverage (Even Parity):

p1 (Bit 0 in pos is 1) → Checks: 1, 3, 5, 7 →  $p1 = d1 \oplus d2 \oplus d4$

p2 (Bit 1 in pos is 1) → Checks: 2, 3, 6, 7 →  $p2 = d1 \oplus d3 \oplus d4$

p4 (Bit 2 in pos is 1) → Checks: 4, 5, 6, 7 →  $p4 = d2 \oplus d3 \oplus d4$

Parity bits placed strictly at powers of 2 (1, 2, 4, 8, 16...)

2. Syndrome Decoding Word (S4 S2 S1)

Syndrome Bit Equations at Receiver:

$s1 = b1 \oplus b3 \oplus b5 \oplus b7$

$s2 = b2 \oplus b3 \oplus b6 \oplus b7$  |  $s4 = b4 \oplus b5 \oplus b6 \oplus b7$

Syndrome Interpretation & Correction:

If Syndrome = 000 → No error! Data accepted.

If Syndrome ≠ 000 → Binary value equals the ERROR BIT position!

E.g., Syndrome = 110 (binary 6) → Bit 6 is corrupted!

Correction Action: Simply invert bit 6 (0 → 1 or 1 → 0)!

No retransmission required! Forward Error Correction (FEC).

Hamming Distance Theorem & Redundancy Inequality (AKTU Core Formulas)

Redundancy Bits Formula:

$2^r \geq m + r + 1$

where m = data bits, r = check/parity bits. For m = 4 bits:  $2^3 = 8 \geq 4 + 3 + 1 = 8$  (r = 3).

Hamming Distance  $d_{min}$ :

Number of bit positions in which two codewords differ (XOR weight).

Error Detection Condition:

$d_{min} \geq d + 1$  (To detect up to 'd' bit errors)

Error Correction Condition:

$d_{min} \geq 2t + 1$  (To correct up to 't' bit errors)

## 1. Hamming Distance & FEC

Forward Error Correction allows the receiving terminal to detect and automatically correct bit flips without requiring retransmission. A code with minimum distance  $d_{min}$  can detect  $d$  errors if  $d_{min} \geq d + 1$  and correct  $t$  errors if  $d_{min} \geq 2t + 1$ .

## 2. Hamming (7, 4) Syndrome Decoding

## Syndrome Word Error Locator

$$S1 = b1 \oplus b3 \oplus b5 \oplus b7$$

$$S2 = b2 \oplus b3 \oplus b6 \oplus b7$$

$$S4 = b4 \oplus b5 \oplus b6 \oplus b7$$

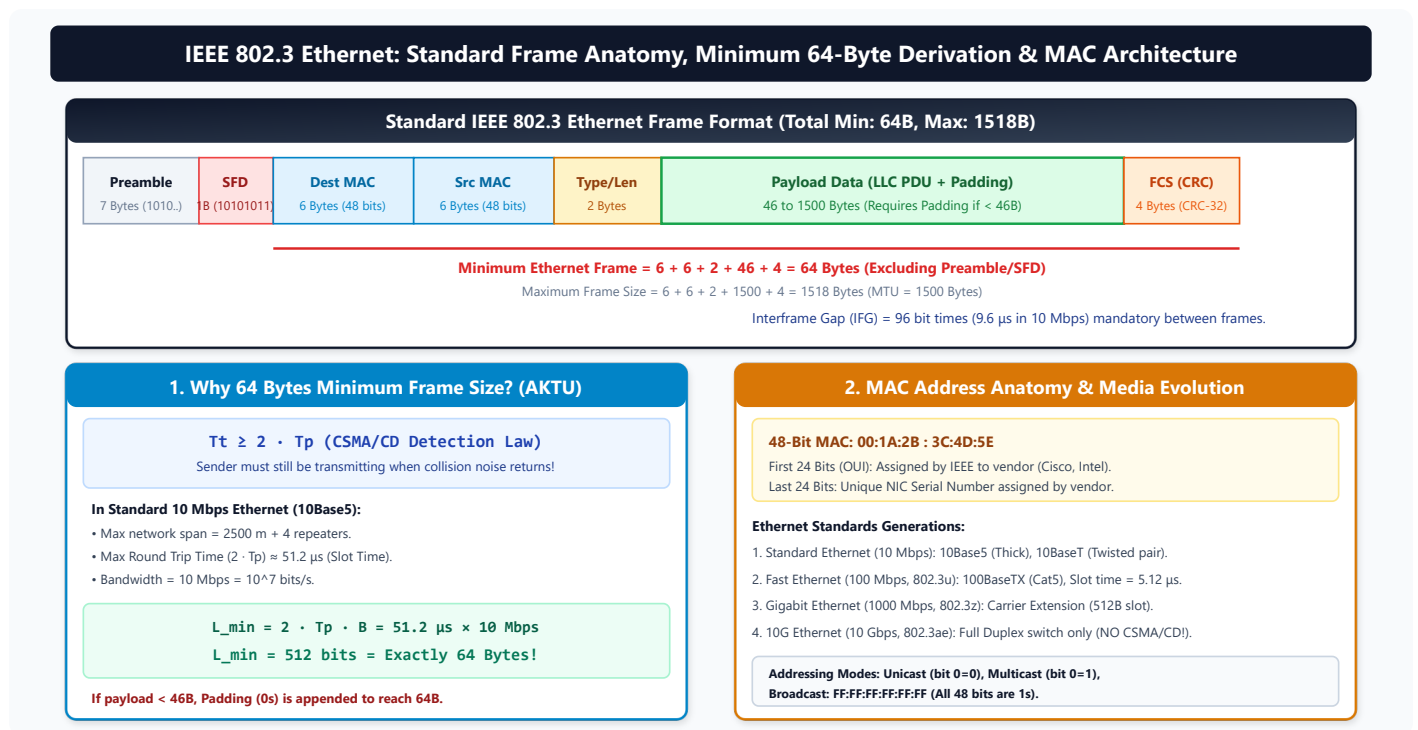
$$\text{Syndrome } (S4 \ S2 \ S1)_2 = \text{Error Bit Index (1 to 7)}$$

# Module 12: IEEE 802 LAN Standards & Ethernet (IEEE 802.3)

## Unit 2: Data Link Layer & MAC Sublayer

## IEEE 802 Architecture & Ethernet Evolution

LLC 802.2 vs MAC decomposition, complete IEEE 802.3 frame structure, formal mathematical derivation of the 64-byte minimum frame size, and 48-bit MAC addressing.



## 1. IEEE 802 Architecture & Frame Anatomy

The IEEE 802 architecture decouples transmission media specifics into the MAC sublayer while abstracting logical flow and network-layer multiplexing into the LLC 802.2 sublayer. The standard Ethernet frame spans between 64 and 1518 bytes.

## 2. 64-Byte Minimum Frame Size Derivation

## Collision Detection Slot Time

Worst-case Round Trip Time ( $2 \cdot T_p$ ) = 51.2  $\mu$ s

$L_{\min} = (2 \cdot T_p) \cdot \text{Bandwidth} = 51.2 \mu\text{s} \cdot 10 \text{ Mbps} = 512 \text{ bits} = 64 \text{ Bytes}$

Minimum Payload = 64B - 18B (Header+Trailer) = 46 Bytes

# Module 13: Token Bus (802.4), Token Ring (802.5) & FDDI Architecture

Unit 2: Data Link Layer & MAC Sublayer

Deterministic LANs & Optical Self-Healing

Comprehensive architectural study of IEEE 802.5 Token Ring, MSAU star-ring wiring, Active Monitor recovery, and FDDI dual counter-rotating ring self-healing loopback mechanics.

Token Ring (IEEE 802.5) & FDDI Dual Counter-Rotating Self-Healing Architecture

1. IEEE 802.5 Token Ring & MSAU

**Token Format (3B):**  
SD (1B) + AC (1B) + ED (1B)

**Active Monitor (S1):**

- Prevents orphaned frames.
- Regenerates lost token.
- Sets Monitor bit in AC.

**Speed: 4 / 16 Mbps**

**Early Token Release (ETR):**  
In 16 Mbps mode, station releases token immediately after transmitting frame without waiting for frame header to travel around the entire ring.

2. FDDI Dual Counter-Rotating Self-Healing Ring

**Self-Healing Wrap:**  
When cable snaps, stations B and C wrap primary ring into secondary ring!  
**Forms 1 Big Single Ring**  
Network stays 100% alive!  
**Span: 200 km, 1000 Stns**

**Timed Token Protocol in FDDI:**  
TTRT (Target Token Rotation Time) negotiated by stations.  
Token Holding Time:  $THT = TTRT - TRT$  (Guarantees Real-time voice bandwidth!).

Comparative Architecture: Token Bus (802.4) vs Token Ring (802.5) vs FDDI

|                            |                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Token Bus (802.4):</b>  | Physical Bus, Logical Ring. Stations ordered by descending numerical address. Complex ring maintenance.                     |
| <b>Token Ring (802.5):</b> | Physical Star / Logical Ring via MSAU. Token captured by setting T bit in Access Control byte. Monitored by Active Monitor. |
| <b>FDDI (ANSI X3T9.5):</b> | Dual Counter-Rotating Fiber Rings. 100 Mbps. 4B/5B block coding. Self-healing wrap-around on fiber rupture.                 |
| <b>Efficiency Law:</b>     | FDDI Token Holding Time: $THT = \max(0, TTRT - TRT)$   Efficiency $\eta = THT / TTRT$                                       |

## 1. Token Passing & Active Monitor

In Token Ring (IEEE 802.5), stations utilize a 3-byte circulating token frame. An Active Monitor prevents network deadlock by eliminating orphaned circulating frames via the Monitor Bit and regenerating lost tokens.

## 2. FDDI Dual Ring Self-Healing

## FDDI Timed Token Equations

$$THT = \max(0, TTRT - TRT)$$

$$\text{Transmitted Asynchronous Bits} = THT \cdot 100 \text{ Mbps}$$

Upon a cable cut, adjacent Dual Attachment Stations (DAS) execute optical wrap-around, collapsing the primary and secondary counter-rotating rings into a single fault-tolerant operational ring.

# Module 14: Bridges, Switches, Spanning Tree Protocol (STP) & Unit 2 AKTU PYQs

Unit 2: Data Link Layer & MAC Sublayer

Layer 2 Switching & AKTU Exam Archive

Deep-dive into transparent bridge MAC address learning, filtering, and flooding, bridging loops, IEEE 802.1D Spanning Tree Protocol root election, and verified model solutions for AKTU BCS603 exam questions.

## Transparent Learning Bridges & Spanning Tree Protocol (IEEE 802.1D STP)

### 1. Transparent Bridge Learning Algorithm



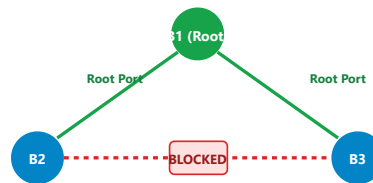
#### • The Three Forwarding Rules:

1. **LEARN:** Record Source MAC + Arrival Port in Forwarding Database.
2. **FORWARD / FILTER:** If Dest MAC is known on another port → Send ONLY there. If Dest MAC is on the SAME port as source → FILTER (Drop frame, save link!).
3. **FLOOD:** If Dest MAC is unknown or Broadcast → Forward to ALL ports except arrival!

#### Collision Domain vs Broadcast Domain:

Bridge/Switch breaks Collision Domains into 1 per port. Maintains 1 single Broadcast Domain.

### 2. Spanning Tree Protocol (STP 802.1D)



#### • The 4-Step STP Algorithm:

1. **Elect Root Bridge:** Bridge with Lowest Bridge ID (Priority + MAC).
2. **Select Root Ports (RP):** Port on each non-root bridge with lowest path cost to Root.
3. **Select Designated Ports (DP):** Port per LAN segment with lowest path cost to Root.
4. **Block Alternate Ports:** Put remaining loop-causing ports in **BLOCKING** state!

## AKTU BCS603 Unit 2 Core Previous Year Exam Questions (10 Marks Distribution)

- |                                     |                                                                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1 (Framing &amp; Stuffing):</b> | Explain Bit Stuffing (5 ones rule) and Byte Stuffing (ESC character). Given bit stream, apply bit stuffing.                          |
| <b>Q2 (ALOHA &amp; CSMA/CD):</b>    | Derive Throughput of Pure and Slotted ALOHA. Prove $T_t \geq 2T_p$ condition in CSMA/CD and 64-byte frame size.                      |
| <b>Q3 (Flow Control ARQ):</b>       | Derive Stop-and-Wait efficiency $\eta = 1/(1+2a)$ . Prove why $W_s = 2^m - 1$ in GBN and $W_s = 2^{m-1}$ in Selective Repeat.        |
| <b>Q4 (Error &amp; STP):</b>        | <b>Solved CRC numerical with polynomial <math>G(x)</math>, Hamming (7, 4) syndrome calculation, and STP root election algorithm.</b> |

## 1. Transparent Bridge Operation

Layer 2 switches partition collision domains on a per-port basis while preserving a single broadcast domain. Forwarding tables dynamically learn source MAC addresses and intelligently filter or forward frames, eliminating unnecessary collisions.

## 2. Spanning Tree Protocol (STP 802.1D)

## STP Algorithmic Steps

1. Elect Root Bridge (Lowest Bridge ID = Priority + MAC)
2. Select Root Port on non-root bridges (Lowest path cost to Root)
3. Select Designated Port per segment (Lowest path cost to Root)
4. Place remaining redundant loop-causing ports in BLOCKING state